SHRUTHI KANNAN

| Boston, Massachusetts | 02130 | +1 857 340-3593
| kannan.shru@northeastern.edu |
| https://www.linkedin.com/in/shruthi-kannan-69924624a/ |
| https://github.com/sk28352 |

EDUCATION

Northeastern University
Master of Science, Computer Science 2025 – 2026
Gpa: 3.333

SRM Institute of Science and Technology (SRMIST), Tamil Nadu 2020 – 2024
Bachelor of Technology, Computer Science and Engineering

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, C++, C
Databases: MySQL, MongoDB, ChromaDB (Vector DB)
Predictive Modeling & Time-Series Forecasting: ARIMA, SARIMA, Prophet, LSTM, Gradient Boosting (LGBM, XGBoost), Ensemble Methods (Random Forest, AdaBoost, Stacking Regressors), Regression Analysis
Machine Learning & Deep Learning: Scikit-learn, PyTorch, TensorFlow, Keras, Model Evaluation & Cross-Validation
NLP & LLMs: RoBERTa, Sentence-BERT, LLaMA 3 (via Ollama), LangChain, ChromaDB, RAG Pipelines, Emotion/Intent Classification
Computer Vision: OpenCV, MobileNet, CNN architectures
Data & Analysis: Pandas, NumPy, Data Preprocessing, Exploratory Data Analysis, Statistical Analysis
Tools: Git, GitHub

PUBLICATIONS

Human Gait Recognition using Cross View Micro Gait Dataset with Light Weight MobileNet (https://ieeexplore.ieee.org/document/10134510) May 2023

- Cross View Micro Gait (CVM-GAIT) Dataset was created with numerous individual recorded videos in the cross view with various speeds, which was converted into frames and stored as images.
- This study was carried out with SVM, Decision tree, Inception net and the proposed Lightweight Mobile net architecture.
- MobileNet V2 achieved the best performance with **98.9%** training accuracy and **98.7%** test accuracy, outperforming Inception V3 and traditional classifiers in precision, recall, and F1-score.

Solar Power Output Prediction with AI: Leveraging Meta Learner Ensemble Stacked Regressor, (https://ieeexplore.ieee.org/document/10939884) March 2025

- Developed an ensemble model combining K-Nearest Neighbors (KNN), Deep Neural Networks (DNN), Light Gradient Boosting Machine (LGBM), and Random Forest with a Meta-Learner via Scikit-learn's stacking regressor.
- Fine-tuned models through hyperparameter tuning to capture non-linear relationships.
- Improved forecasting accuracy, achieving an **R^2 score of 0.68**, strengthening reliability in renewable energy prediction.

PROJECTS

Northeastern University RAG-Powered Student Support Chatbot Aug 2025

- Built a Retrieval-augmented generation chatbot using LangChain and a locally hosted LLaMA 3.2 to answer international student queries across 8 categories (visa, employment, billing, etc.) using official university documents as the knowledge source.
- A document ingestion pipeline was made using PyPDF for parsing, persistent vector storage in ChromaDB with category-based filtering to improve retrieval precision and avoid redundant reprocessing.
- Tuned generation for factual accuracy (temperature = 0.2), added transparent source citations for every response, achieving sub-30-second response times with fully source-traceable answers in testing.

Hybrid Transformer Based Mental Health Chatbot Feb 2025 – April 2025

- Engineered an end-to-end empathetic chatbot pipeline combining RoBERTa (multi-label emotion detection), Sentence-BERT (intent recognition) and LLaMA 3 (generative responses via Ollama).
- Achieved 0.95 emotion-detection accuracy and 0.85 intent-classification accuracy on real-world mental health dialogues, with average empathy & relevance scores of 1.00 and 0.55, respectively.
- Benchmarked against SVM, Naïve Bayes and LSTM baselines to validate superior balance of emotional understanding and response relevance.

Wind Energy and Turbine Power Prediction using Machine Learning Dec 2023 – June 2024

- Built forecasting pipelines on meteorological and wind turbine datasets to predict power generation.
- Time-series models like Auto Regressive Integrated Moving Average (ARIMA), SARIMA were used and compared with Prophet, LSTM (Long Short-Term Memory) and a fine-tuned gradient model.
- On turbine power data models like Linear Regression, Bayesian Analysis, KNN, MLP (Multi-layer perceptron), SVR (Support vector Regression), Random Forest with MLP Regressor having R^2 0.99 and RMSE = 1.58 kW were used.